Why sourdough rises

Sourdough rises because wild yeast and lactic acid bacteria in the starter feed on the sugars in flour and give off carbon dioxide gas. The gas gets trapped in the dough and makes it expand.

The trapping is done by gluten, a stretchy network of proteins that forms when flour and water are mixed and kneaded. Folding the dough a few times during the first rise strengthens that network so it can hold more bubbles.

The bacteria also make acids as they work, which gives sourdough its tangy flavour and helps the loaf keep longer. A warm kitchen speeds up fermentation, and a cold fridge slows it down and builds more flavour overnight.

In the oven, the trapped gas expands once more and the loaf jumps up, called oven spring. Baking with steam keeps the crust soft long enough for that final rise.